Brittle Star-Inspired Damage Robustness: Multi-Agent Reinforcement Learning Approach

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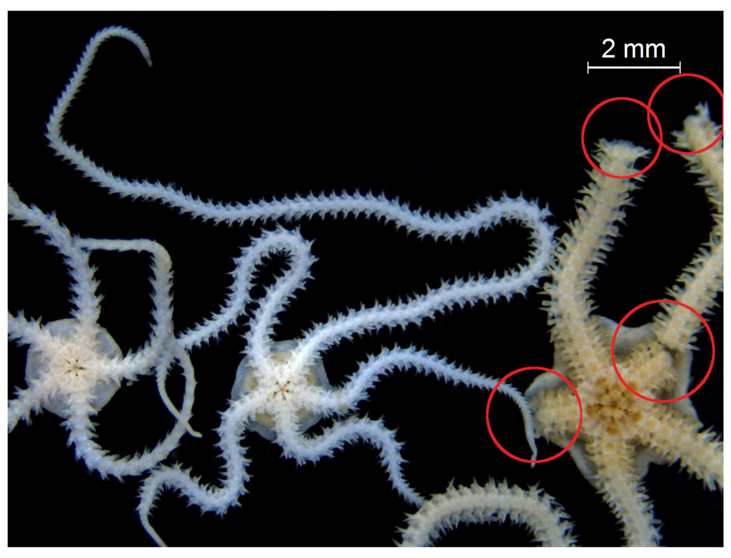
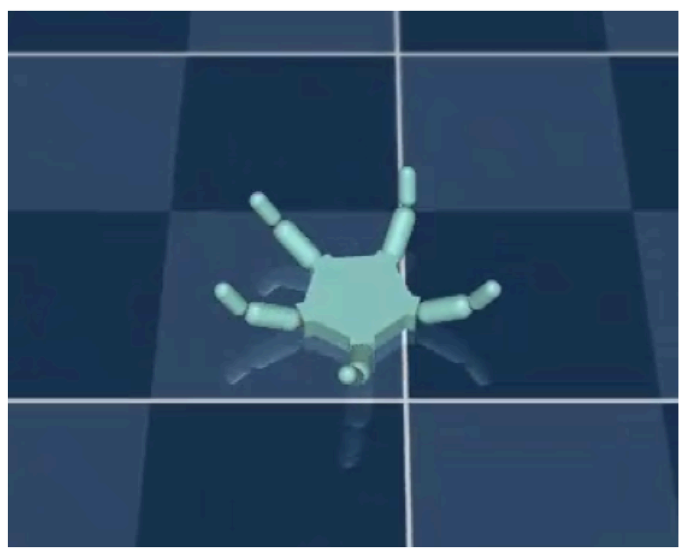
1 Damage Robustness

Real-world robots might sustain damage in operation. Nature already has solutions to this problem, e.g. brittle stars:



Two intact (left) and one damaged (right) brittle stars.



Brittle star-inspired robot with 5 arms in a simulated environment.

brittle star damage robustness $\xrightarrow{?}$ damage-robust robots

2 Research Goal

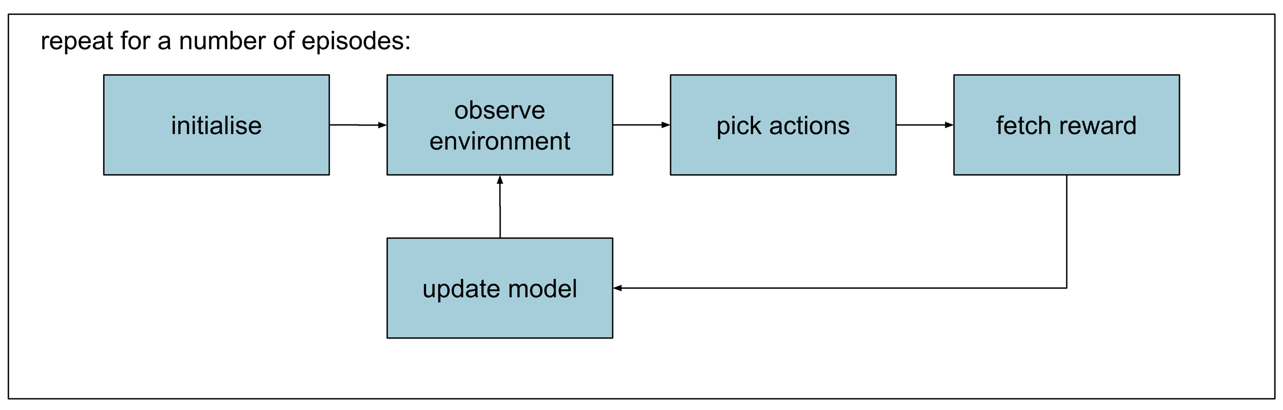
Build damage-robust robots by learning brittle star-inspired locomotion gaits with the help of central pattern generators (CPGs) and multi-agent reinforcement learning (MARL).

3 Methodology

- actions: rowing gait modulation through CPGs:

action	description
leading arm	pointing upward
left rower	clockwise rotation
right rower	counterclockwise rotation
secondary left rower	smaller amplitude clockwise
secondary right rower	smaller amplitude counterclockwise

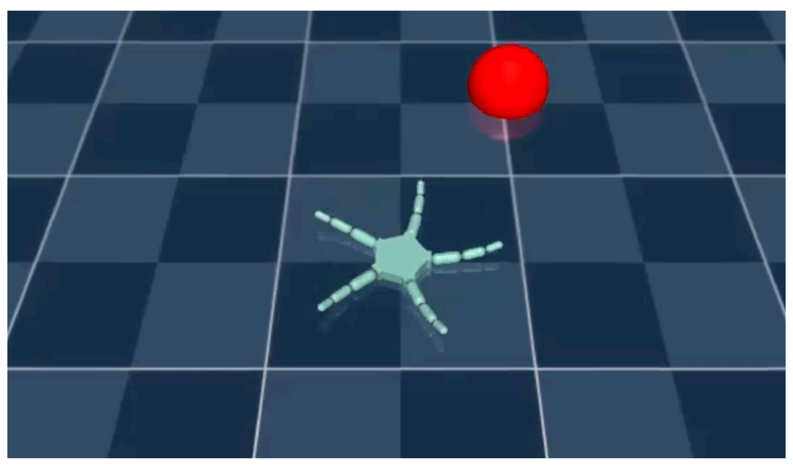
- learning: **action selection** through MARL:
  - observations:
    1. **angle** between arm and target,
    2. **direction** between body and target,
    3. **distance** between body and target
  - reward function: **distance reduction** toward target
  - **fixed** target



Overview of the methodology.

4 Results

Can walk in simulation with 5 arms, fixed target and no damage:



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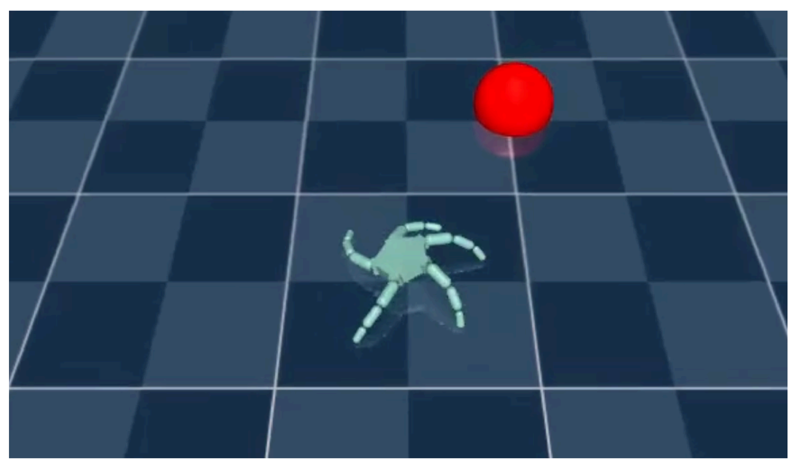

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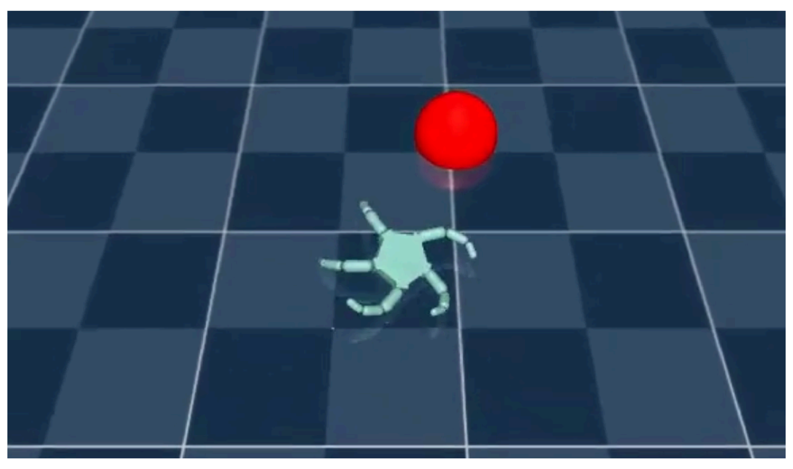

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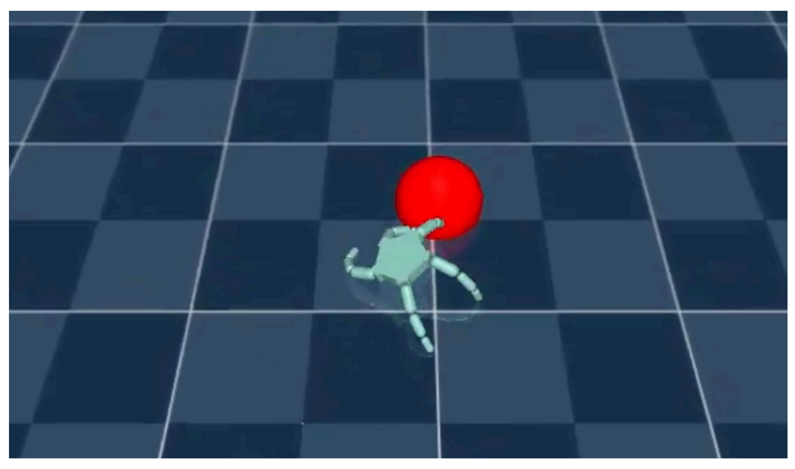
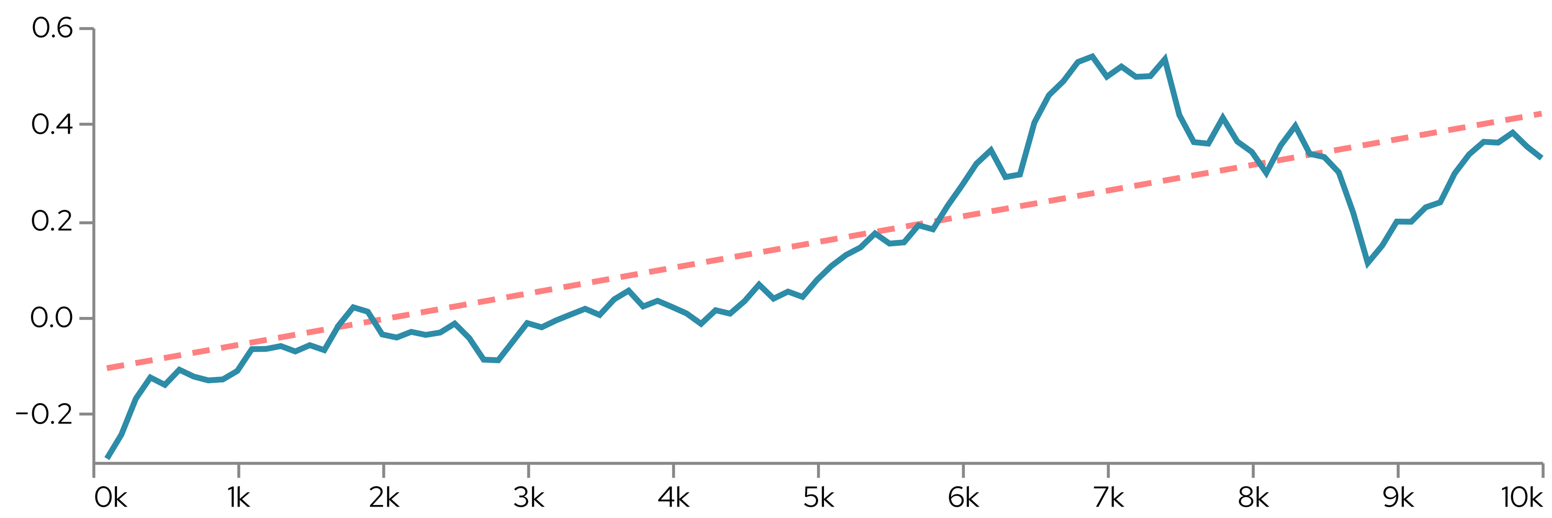


4 / 4

However, this did not reflect in high rewards during training:



average reward over 10 environment steps while training for 100 episodes

5 Conclusions

6 Acknowledgements

7 References

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